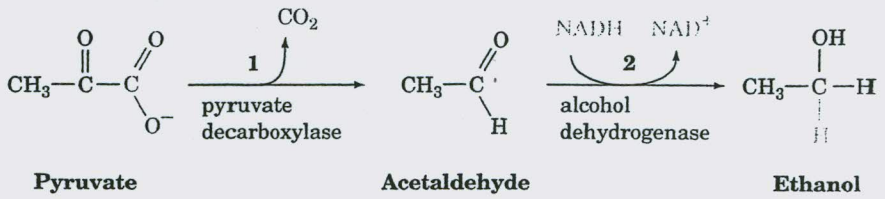
Biochemistry Cume
March 31, 2006

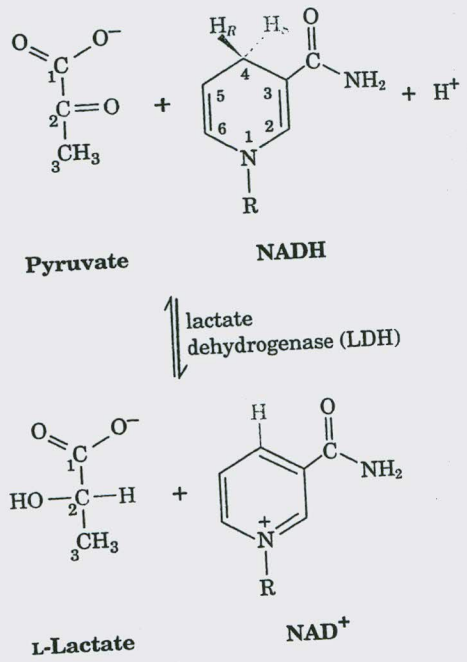
1. Brewers yeast cultures can grow with glucose as a carbon and energy source either aerobically or anaerobcially. Discuss the differences in these two modes of growth including the ultimate fate of carbon and the energy yield.

2. If the brewers yeast cells were grown in the presence of glucose labeled with C-14 at the number 3 carbon, in what molecule would the radioactive carbon end up in each case? How would the results differ in the C-14 label were at carbon 4 of glucose?

3. A mutant has been observed in an enzyme of the glycolysis pathway which makes that enzyme totally inactive. This mutant dies if the brewers yeast is grown on glucose under anaerobic conditions. If grown under aerobic conditions, the brewers yeast cultures survive, but grow and divide at a much slower rate (about half the normal rate). What is the most likely enzyme or reaction affected by this mutation? Explain how the mutation results in the differences in growth behavior relative to wild type (normal) yeast cells.

4. If the experiment described in question 2 were repeated with this mutant, would the results be the same or different? Why?


CO2
1
pyruvate
decarboxylase
NADH
NAD+
2
alcohol
dehydrogenase
Pyruvate
Acetaldehyde
Ethanol


Pyruvate
NADH
lactate
dehydrogenase (LDH)
L-Lactate
NAD+

B-3

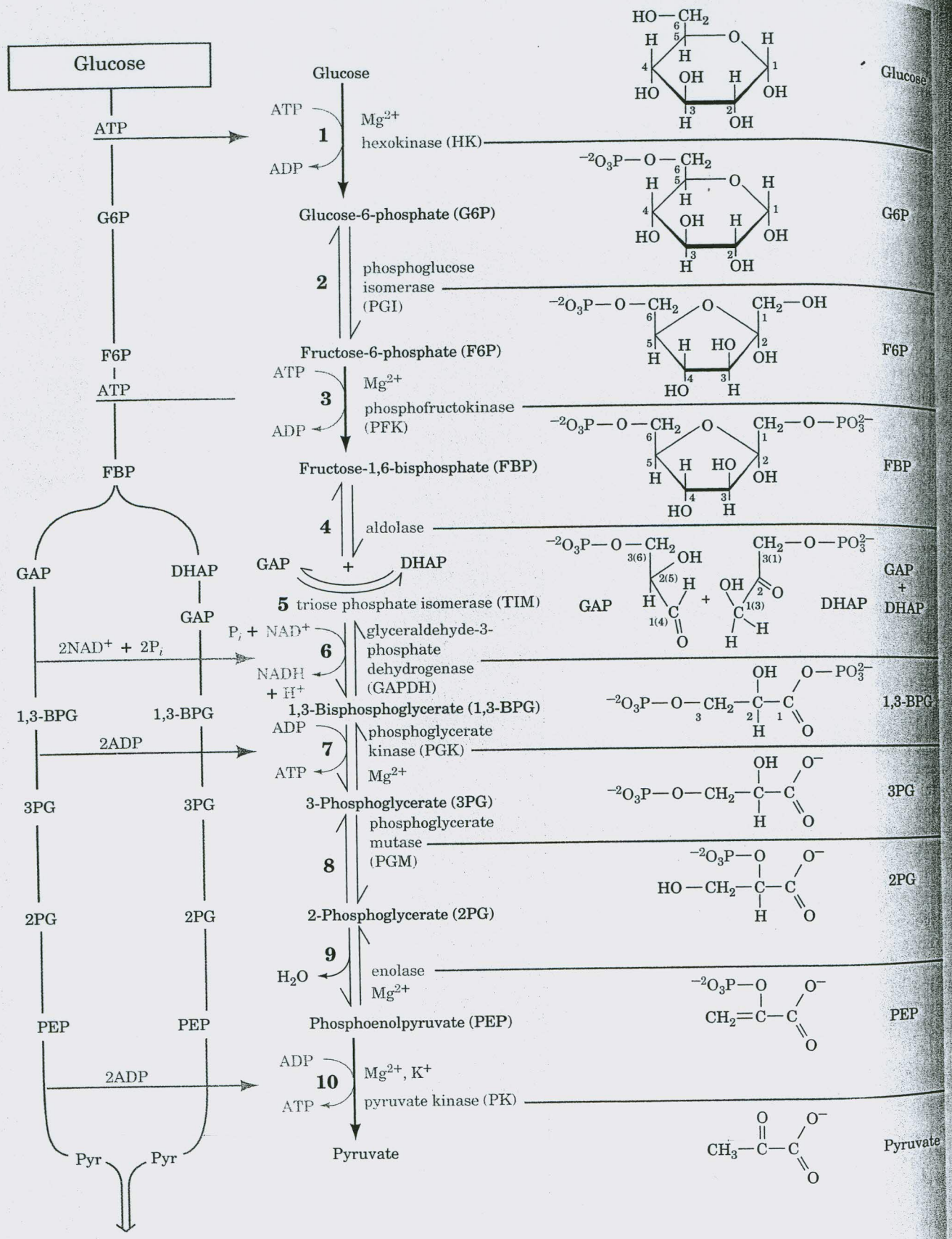


FIGURE 17-3 Degradation of glucose via the glycolytic pathway. Glycolysis may be considered to occur in two stages. Stage I (Reactions **1–5**): Glucose is phosphorylated and cleaved to form two molecules of the triose glyceraldehyde-3-phosphate. This requires the expenditure of two ATPs in an "energy investment" (Reactions **1** and **3**). Stage II (Reactions **6–10**): The two molecules of glyceraldehyde-3-phosphate are converted to pyruvate with the concomitant generation of four ATPs (Reactions **7** and **10**). See the Animated Figures

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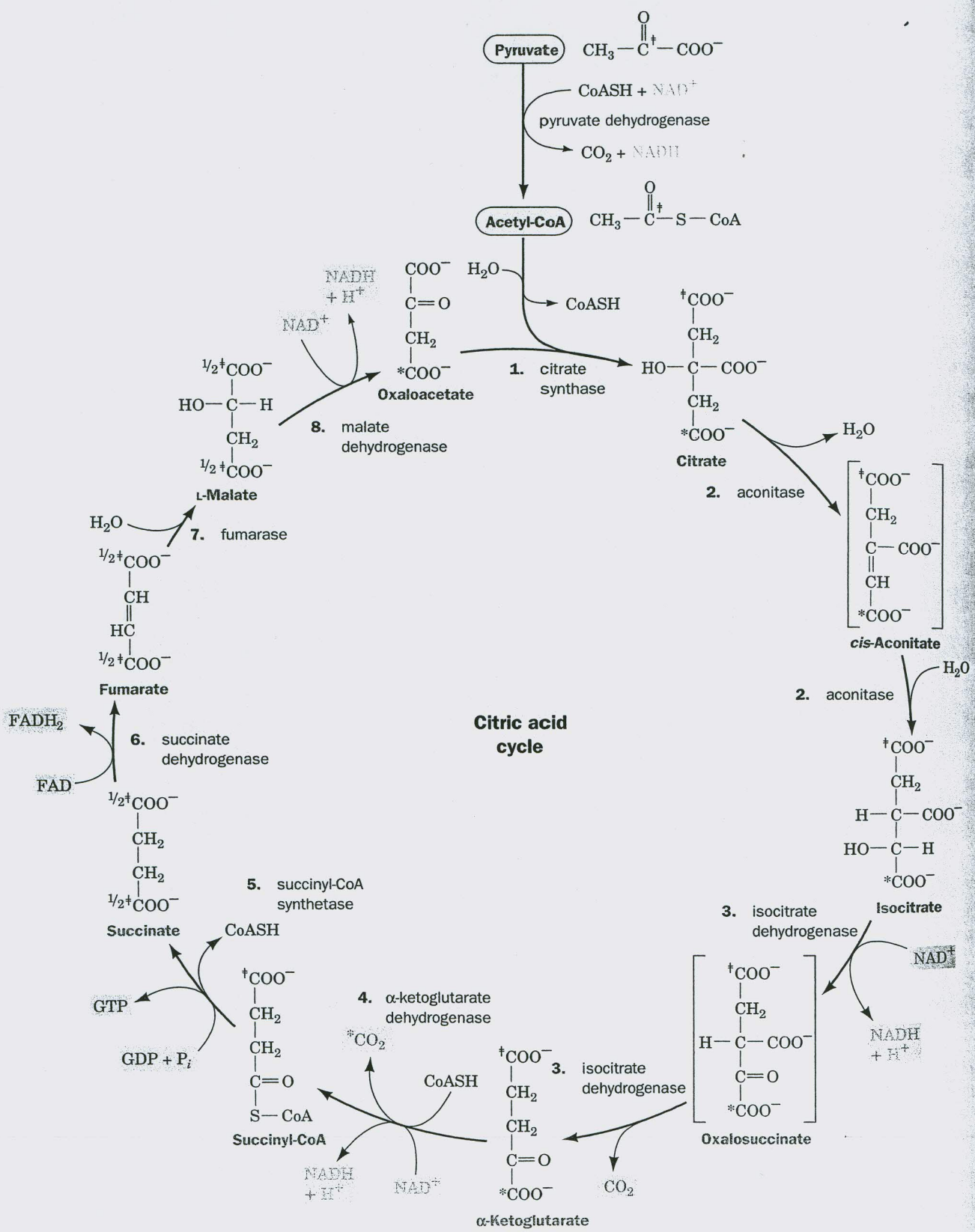

FIGURE 21-1 Reactions of the citric acid cycle. The reactants and products of this catalytic cycle are boxed. The pyruvate → acetyl-CoA reaction (*top*) supplies the cycle's substrate via carbohydrate metabolism but is not considered to be part of the cycle. The bracketed compounds are enzyme-bound intermediates. An isotopic label at C4 of oxaloacetate (*) becomes C1 of α-ketoglutarate and is released as CO_2 in Reaction 4. An isotopic label at C1 of acetyl-CoA (‡) becomes C5 of α-ketoglutarate and is scrambled in Reaction 5 between C1 and C4 of succinate (1/2‡). See the Animated Figures

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